

Motorized Pedal Boat Project

Interdisciplinary Embedded Systems & Motor Design Project

Role: Lead Controls Engineer | Team: 6 engineers (4 ME, 1 EE, 1 CE)

github.com/NotBryson4Real/Motorized-Pedal-Boat

Overview

A motorized pedal boat built during my junior year of college as a cross-disciplinary project. The challenge was to take a standard pedal boat with its pedaling mechanism removed and engineer a complete propulsion, steering, and controls system capable of navigating a timed obstacle course and competing in head-to-head races. Our team designed and hand-wound custom BLDC motors, fabricated the drivetrain, and built all controls and UI from scratch, all within a \$600 budget.

My Contributions

- Sole developer for all firmware across both microcontrollers (TI MSP430 for motor controls & Arduino Mega for UI/instrumentation)
- Designed and implemented a dual-axis joystick control system on the MSP430, translating analog joystick input into differential-drive PWM signals for two custom BLDC motors
- Built a complete driver UI system on the Arduino Mega featuring three I2C LCD screens displaying real-time battery voltage, battery temperature, and boat system status
- Designed analog sensing circuits including a voltage divider for battery voltage monitoring and a thermistor-based voltage divider for battery temperature sensing
- Resolved a critical logic-level mismatch between the MSP430's 3.3V PWM output and the ESCs' 5V logic requirement to ensure reliable motor control signals
- Contributed to full boat wiring, including LED accent lighting and integration of water-cooled ESCs using aquarium pumps to solve ESC overheating issues
- Assisted with motor winding alongside the full team, helping overcome dielectric failure challenges during fabrication of the custom BLDC motors

Technical Details

Hardware: TI MSP430 (motor controls), Arduino Mega (UI/instrumentation), custom hand-wound BLDC motors, ESCs with aquarium-pump water cooling, dual-axis 10kΩ potentiometer joystick, 3× I2C LCD screens, thermistors, voltage divider circuits, WS2812B LED strip, Ryobi 18V 4Ah battery packs

Communication: I2C (LCD screens & sensors), analog signaling (joystick to MSP430), PWM (ESC control)

Software: Embedded C/C++ on both platforms, differential-drive mixing algorithm for joystick-to-motor translation, real-time sensor polling and display refresh across three LCD screens

Key Challenge — Logic Level Mismatch: The MSP430 outputs 3.3V PWM signals, but the ESCs required 5V logic. This mismatch caused unreliable motor behavior until we identified the root cause and implemented level shifting to bridge the gap.

Key Challenge — Thermal Management: ESCs were overheating under sustained load. We engineered a water-cooling solution using aquarium pumps to circulate pool water through the ESC enclosures, keeping temperatures within safe operating range during extended runs.

System Architecture

Circuit 1 — Motor Controls (MSP430): Reads dual-axis joystick, computes differential-drive mixing, outputs PWM to two ESCs driving the custom BLDC motors for forward/reverse and left/right steering.

Circuit 2 — UI & Instrumentation (Arduino Mega): Manages three I2C LCD screens, reads battery voltage via voltage divider, reads battery temperature via thermistor voltage divider, and drives LED accent lighting.

Project Photos

